INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Business Continuity and Disaster Recovery Assignment

GRC205 Incident Management and Business Continuity

Student name	OUABO KAMDEM DERRICK ARNAUD
Student ID	2025/GCR/12778
Official email	gcr2512778@students.icdfa.edu.ng
Programme	CGRCE
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Introduction, role and assumptions

I act as the Business Continuity and Disaster Recovery (BC/DR) Coordinator for Northbridge Skills Institute, a fictional provider with 1,800 learners and 90 staff. At 08:00 WAT on Day 0 (a Monday), flooding and a power failure made the main site and its local systems unavailable, and Facilities expects the site to stay inaccessible for 72 hours. No cyberattack is confirmed. This report presents the six required tasks in order, using the supplied case data, the ICDFA teaching deck (slides 1 to 15) and the official guidance listed in the reference list.

All times are WAT and elapsed times run from 08:00 on Day 0. Slide numbers follow the order of the teaching deck, with the maturity model on slide 13. Costs are in Nigerian naira (NGN) and are exercise estimates, not observed losses. Schedule results are a modelled baseline, not a live test. Where the case does not supply a fact, I label it as an assumption in Appendix A (A1 to A9). The alignment with NIST SP 800-34 and ISO 22301 is educational and does not claim certification or compliance.

Task 1: Business Impact Analysis

The BIA is the foundation of every later decision (ICDFA, slide 5; NIST, 2010). Table 1 records the five functions in the structure from slide 5: critical functions, dependencies, resources and users. Table 2 sets out the impact over time, the minimum acceptable service (the lowest workable service plus the approval and record controls any manual process needs) and the recovery priority.

Table 1: Business Impact Analysis: function profile, resources and dependencies

Function	Owner role	Users affected	Critical resources (people, premises, suppliers, information)	Dependencies
F1 Teaching and assessment (LMS01, DB01)	Head of Academic Delivery	1,800 learners; 120 candidates at 11:00; teaching staff and invigilators (number not supplied, A1)	People: Exams Officer, invigilators, one technician. Premises: alternate facility (25 workstations from 10:00). Suppliers: Microsoft 365, alternate-facility provider. Information: approved exam papers, candidate list, access arrangements.	Recovery network, AUTH01, DB01, LMS01; staff, secure access, communication
F2 Fees and finance (FIN01, DB01)	Finance Manager	1,800 learners and payers; finance staff	People: finance officers (maker and checker). Suppliers: payment provider (available), bank. Information: provider transaction reports, wallet balances, receipt sequence.	Network, AUTH01, DB01, FIN01; provider reports
F3 Admissions and registration (REG01, DB01)	Admissions and Registry Manager	Applicants (number not supplied, A1); registry staff	People: admissions staff. Premises/channel: temporary admissions mailbox and form. Information: application forms, supporting documents, reference numbers.	Network, AUTH01, DB01, REG01
F4 Staff payroll (HR01, DB01)	HR and Payroll Manager	90 staff	People: payroll officer and approver. Suppliers: bank submission portal. Information: attendance and variation data, last approved payroll file.	Network, AUTH01, DB01, HR01; bank deadline
F5 Records and certificates (FS01)	Registrar	Learners, alumni and staff requesting records (number not supplied, A1); registry staff	People: records officer. Information: certificate templates, access-restriction list, request log.	Network, AUTH01, FS01 (no DB01 needed)

Table 2: Impact over time, minimum acceptable service and recovery priority

Function	Impact over time	Minimum acceptable service and controls	Recovery priority
F1	NGN 60,000/h from hour 0. Exam at 11:00 (hour 3): if no acceptable assessment service exists, NGN 120,000 once. Learner access and assessment integrity at risk.	Secure Microsoft 365 assessment for 105 candidates at 11:00 (25 centre workstations + 80 own devices); the other 15 sit later the same day. Controls: Academic Lead approves the alternate paper; photo-ID check; timed, logged submissions; numbered mark sheet.	1 (business). Technical: after DB01
F2	NGN 40,000/h. Fee confirmation due 16:00 (hour 8). Local receipts and wallet records cannot update; reconciliation needed.	Accept payments through the provider; issue numbered manual receipts; maker-checker log; confirm fees from the provider report.	2 (business). Technical: after DB01
F3	NGN 20,000/h. Admissions close 18:00 (hour 10). New applications and changes cannot be entered locally.	Controlled admissions mailbox and form; register with reference numbers; two-person count at close; deadline kept.	3 (business). Technical: after DB01
F4	NGN 10,000/h. Bank deadline 12:00 Day 1 (hour 28). Wrong or duplicate instructions harm staff and need correction.	No manual run unless HR01 fails; then last approved payroll plus a variance sheet. Dual authorisation (HR and Finance) before bank submission; duplicate check.	4 (business). Technical: after DB01
F5	NGN 5,000/h. 48-hour service commitment. Integrity and access restrictions must hold.	Log requests; issue nothing from unverified copies; Registrar verifies against restored FS01 before release.	5 (business). Technical: after checks, parallel with DB01

Table 3: Disruption cost calculation (no recovery or continuity measures)

Function	Rate (NGN per hour)	8 hours (NGN)	24 hours (NGN)
F1 Teaching and assessment	60,000	8 × 60,000 = 480,000	24 × 60,000 = 1,440,000
F2 Fees and finance	40,000	8 × 40,000 = 320,000	24 × 40,000 = 960,000
F3 Admissions and registration	20,000	8 × 20,000 = 160,000	24 × 20,000 = 480,000
F4 Staff payroll	10,000	8 × 10,000 = 80,000	24 × 10,000 = 240,000
F5 Records and certificates	5,000	8 × 5,000 = 40,000	24 × 5,000 = 120,000
Subtotal of hourly impacts	**135,000**	**1,080,000**	**3,240,000**
Examination rescheduling (one-off, trigger met at 11:00)	–	120,000	120,000
Combined total		**1,200,000**	**3,360,000**

The five rates sum to NGN 135,000 per hour, so eight hours cost 8 × 135,000 = NGN 1,080,000 and 24 hours cost 24 × 135,000 = NGN 3,240,000. The examination starts at hour 3, inside both windows,

so the one-off NGN 120,000 applies once in each case. The combined totals are NGN 1,200,000 and NGN 3,360,000 (Table 3).

These figures have two limitations. First, they apply fixed, linear hourly rates and ignore cost cliffs: a missed payroll bank deadline or admissions closure causes harm that is not proportional to time, and the calculation assumes no mitigation and stops at 24 hours. Second, they are exercise estimates, not observed losses, so they omit recovery labour, reconciliation effort, alternate-facility costs and later-period effects. Financial impact must also be kept separate from learner welfare (stress, missed progression), reputation (trust of learners and sponsors) and data integrity (wrong grades, receipts or pay), none of which money alone repairs.

The 11:00 examination makes F1 the first business priority: it is the earliest hard deadline, it affects 120 learners and it carries both the one-off cost and assessment-integrity risk. The later deadlines then set the order: F2 (16:00 Day 0), F3 (18:00 Day 0), F4 (12:00 Day 1) and F5 (48 hours), which also follows the descending hourly rates. Business priority is not technical restoration order. F1 to F4 all need the recovery network, AUTH01, the security and backup checks and DB01 first, so the technical order is network, AUTH01, checks, DB01, then the four applications. FS01 needs no database and can be restored in parallel with DB01 even though F5 ranks last. Slide 5 makes the same distinction between criticality ranking and recovery sequence.

Five dependency risks matter most. (1) AUTH01 is a single point of failure for every local function and its snapshot is 8 hours old. (2) DB01 supports F1 to F4, and its restore needs an unbroken chain from B01 through every incremental to 07:00, so one bad link hits four functions. (3) Two technicians are a people bottleneck, because each action occupies one technician for its full duration. (4) The production and network backup administrator identity is shared and the provider’s access independence is undocumented, so one compromised credential could reach every copy. (5) The examination relies on papers and candidate lists that I have not verified are available outside the failed systems (A3).

Task 2: Recovery objectives and backup design

2.1 Objectives and targets

Following NIST (2010) and slide 6, the Recovery Time Objective (RTO) is the maximum acceptable time from disruption to restoration of a function. The Recovery Point Objective (RPO) is the maximum acceptable data loss, measured in time before the disruption, so it fixes how recent a restorable checkpoint must be. The Maximum Tolerable Downtime (MTD) is the outage limit beyond which harm becomes irreparable; it includes the RTO plus validation and work-recovery time. Targets are commitments to be justified and tested, not guarantees.

Table 4 sets my targets. Each MTD follows a business limit: 8 hours for F1 and F2, 10 hours for F3 (admissions close at hour 10), 24 hours for F4 (leaving time to validate and submit before 12:00 Day 1) and 48 hours for F5. Each RTO is half the MTD or less, so a buffer remains for validation and acceptance. I declare recovery at Service Level 2: the application or file service is restored, its data checks pass against the expected checkpoint, and the business owner accepts the minimum service. Technical restoration without owner acceptance is not recovery.

Table 4: Recovery objective matrix

Function	RTO	RPO	MTD	Buffer (MTD – RTO)	Justification
F1 Teaching and assessment	4 h	1 h	8 h	4 h	First priority. The 11:00 exam cannot be met by restoration, so the MTD is the end of Day 0 teaching. A 1 h RPO keeps submissions and marks close to the latest checkpoint.
F2 Fees and finance	4 h	1 h	8 h	4 h	Fee confirmation due at hour 8. Provider records allow reconciliation of the gap.
F3 Admissions and registration	5 h	1 h	10 h	5 h	Admissions close at hour 10. Temporary intake holds new applications meanwhile.
F4 Staff payroll	6 h	4 h	24 h	18 h	Payroll data changes slowly and is rechecked; the buffer allows validation before the 12:00 Day 1 bank deadline.
F5 Records and certificates	12 h	8 h	48 h	36 h	48-hour commitment; files change slowly; integrity matters more than speed.
Shared: AUTH01	2 h	4 h	8 h (inherits F1)	6 h	Every function needs it. The baseline restores it by 09:00. A 4 h snapshot cycle limits lost directory changes.

2.2 Checkpoint analysis

Using the 08:00 disruption as the reference, the latest recoverable DB01 point is B02 at 07:00, so up to 1 hour of transactions can be lost. AUTH01 can lose up to 8 hours of directory change and FS01 up to 9 hours of file change (Table 5). The DB01 window equals the RPO for F1 to F3 with no

headroom and meets the F4 RPO. AUTH01 (8 h against 4 h) and FS01 (9 h against 8 h) exceed their RPOs, so these are the gaps the backup design must close.

Table 5: Checkpoint ages at 08:00 Day 0 and comparison with proposed RPOs

Item	Checkpoint (WAT)	Age at 08:00 Day 0	Functions affected	Proposed RPO	Result
DB01: B02 (B01 + incrementals)	07:00 Day 0	08:00 − 07:00 = 1 h	F1 to F4	1 h (F1 to F3); 4 h (F4)	Met; no headroom for F1 to F3
DB01: B01 full (base only)	23:00 Day −1	08:00 − 23:00 (previous day) = 9 h	F1 to F4	n/a	Fallback point if the chain breaks
AUTH01: B03 snapshot	00:00 Day 0	08:00 − 00:00 = 8 h	F1 to F5	4 h	**Gap 4 h** (not met)
FS01: B04 full	23:00 Day −1	08:00 − 23:00 (previous day) = 9 h	F5	8 h	**Gap 1 h** (not met)
Disconnected full backup	17:00 Day −3	72 h − 9 h = 63 h	Last resort, all	n/a	Fallback only; untested

Some lost activity can be rebuilt from temporary records: fee payments from the payment provider's transaction reports, applications and submissions from Microsoft 365 email and forms, and requests from message logs. I cannot assume recoverable any record keyed only into local systems after 07:00 (DB01), any directory change after 00:00 (new accounts, password resets, group changes) or any file saved after 23:00 Day −1. These must be re-created, re-requested or declared lost, and owners must be told which.

2.3 Backup design and verification

Table 6 closes the RPO gaps by shortening backup intervals and applies the 3-2-1 rule (slide 8): three copies (production, local backup disk, encrypted offsite cloud copy), two media types (disk and cloud) and one offsite copy, plus a fourth immutable or offline copy administered separately. To restore DB01 at the current 08:00 point, the exact chain is B01 (full, 23:00 Day −1), then every incremental from 23:00 to 07:00 in order, ending with B02 at 07:00; one missing or failed link stops restoration at the last unbroken point. Under my design, the chain for a Tuesday 08:00 disruption is the Sunday 23:00 full, the Monday 23:00 differential and the four 30-minute incrementals from 06:00 to 07:30.

Table 6: Backup schedule, retention and protection

Item	Full	Incremental / differential	Retention	Offsite and independent copies
DB01	Weekly, Sunday 23:00	Differential nightly 23:00 (Mon to Sat); incrementals every 30 min, 06:00 to 20:00	Incrementals 7 days; daily 35 days; weekly fulls 13 weeks; monthly fulls 12 months (confirm record rules)	Copy 2: local disk. Copy 3: encrypted cloud copy in another region within 1 h. Copy 4: weekly full to immutable storage (lock 35 days) or offline media

Item	Full	Incremental / differential	Retention	Offsite and independent copies
AUTH01	Weekly system-state backup; nightly image	Snapshot every 4 h and after any bulk directory change	Daily 35 days; weekly 13 weeks	Same three offsite and immutable copies; offsite within 1 h
FS01	Weekly, Sunday 23:00	Differential nightly 23:00; incrementals at 06:00, 10:00, 14:00, 18:00	Same as DB01	Same; permissions preserved
All sets	AES-256 encryption; keys held by a separate custodian	Separate backup administrator identity with MFA; no standing delete rights; dual approval for retention changes; alert on every administrator logon	Restore tests: DB01 monthly; AUTH01 and FS01 quarterly	Immutable or offline copy not reachable with production credentials

An offsite copy or live replica alone does not establish recoverability. An offsite copy proves only that data exists: it may be incomplete, corrupt, undecryptable or deletable with the same shared administrator credential. A replica copies corruption and malicious changes within seconds. Only a restore that meets time and data-quality criteria shows recoverability. I therefore check (1) integrity, with hash and catalogue checks on every set, a daily chain-completeness check and a monthly DB01 restore to an isolated environment with a database consistency check; (2) access independence, with separate credentials, a documented provider access model and a test that the production administrator cannot read, alter or delete offsite and immutable copies; and (3) usability, with application smoke tests, record counts compared with last known counts and owner acceptance of sample records.

Unverified assumptions are labelled: that B01 to B04 pass integrity checks (given only for the baseline drill); that the incremental chain is contiguous; that the provider's copies are independent of our administrator identity; and that the 17:00 Day −3 backup, now 63 hours old, can be restored at all. I treat it as a last-resort fallback (A4, A5).

Task 3: Recovery strategy and budget

I compared the three packages with my objectives, staffing, geography, supplier dependency and the risk of propagating corruption (Table 7). The quoted recovery windows are planning estimates that I must validate, so I treat them as claims, not facts.

Table 7: Option matrix: Cold, Warm and Hot packages

Criterion	Cold	Warm	Hot
First critical service (estimate)	24 to 48 h	3 to 5 h (drill: 3 h 45 min)	About 1 h
Data-loss window (estimate)	Up to 24 h	1 to 6 h	Up to 15 min
Meets RTOs: F1 and F2 (4 h)	No	Likely; drill gives 3 h 45 min and 3 h 30 min	Yes
Meets RTOs: F3 (5 h), F4 (6 h), F5 (12 h)	No	Yes (drill: 4 h 15 min, 4 h 15 min, 2 h 30 min)	Yes
Meets 1 h RPO for F1 to F3	No	Only at the low end of the window; needs the 30-minute backup design	Yes
Staffing	Rebuild effort exceeds two technicians	Two technicians suffice (drill)	Needs continuous monitoring and failover skills; two technicians are thin
Geography and supplier	Site must be outside our flood zone and grid; one supplier	Same	Same, with the highest supplier lock-in
Corruption propagation	Low: restores older data	Moderate: restores from backups, so independent copies are essential	High: real-time replication copies corruption and malicious changes
Fits NGN 3,000,000 limit	Yes	Yes	Yes, but exactly at the limit (A9)

Table 8: Itemised annual budget (NGN)

Item	Cold	Warm	Hot
Annual base cost	600,000	1,500,000	2,400,000
Independent protected backups (required addition)	400,000	400,000	400,000
Training and tests (required addition)	200,000	200,000	200,000
Complete annual cost	**1,200,000**	**2,100,000**	**3,000,000**
Headroom under NGN 3,000,000	1,800,000	900,000	0

I recommend the **Warm** package at NGN 2,100,000 per year. It is the only option that meets the F1 to F5 RTOs with evidence from the drill, it leaves NGN 900,000 of headroom, and two technicians

can operate it. Cold cannot restore the assessment service inside an 8-hour MTD. Hot fits the limit exactly, leaves no contingency, depends on skills the institute lacks, and its replication would copy corruption or malicious changes, so independent backups would still be needed. Warm cannot meet three things: a 15-minute RPO (not required by the business); an estimated 1 to 6 hour data-loss window that can exceed my 1-hour RPO unless the 30-minute incremental design is validated; and the 11:00 examination, because the earliest assessment restoration is 11:45 and needs a continuity workaround.

Table 9: Residual risks and acceptance

Residual risk	Accepted by	Treatment and review
Exam at 11:00 cannot be served by restoration	Principal (Executive Sponsor), advised by Head of Academic Delivery	Continuity workaround; pre-approved assessment pack; review after each exercise
Data loss of 1 to 6 h until the design is validated	Finance Manager and Registrar, with Executive Sponsor	30-minute incrementals; provider reconciliation; test measures actual RPO
Supplier timings are unverified estimates	Principal, on Head of IT advice	Validation drill on supplier environment before contract signature
Backup independence is unverified	Executive Sponsor (treatment owner: Head of IT)	Corrective actions 1 and 2 (Table 17); review quarterly

The 24-hour estimate (NGN 3,360,000) exceeds a year of Warm (NGN 2,100,000), while the 8-hour estimate (NGN 1,200,000) does not. That comparison cannot by itself establish annual return on investment. ROI needs the expected annual loss avoided, which depends on the probability of site-loss events, on the downtime each option would really produce, and on observed costs; the estimates are single-outage figures that assume no mitigation. As an illustration only, Warm costs NGN 900,000 more than Cold. If it avoids at least 19 hours of disruption (24 h less 5 h) at NGN 135,000 per hour, it saves about NGN 2,565,000 per event, so it breaks even at roughly one qualifying event every three years, a frequency I cannot yet support. Before approving spend or accepting supplier claims I need: timed restores on the supplier’s environment with our data, written service levels with remedies, evidence that the site is outside our flood zone, references from similar clients, documented backup access independence, and a local risk assessment supporting the event frequency.

Task 4: Business continuity and disaster recovery plan

4.1 Scope, activation and team coordination

The plan covers the main site, F1 to F5, the local systems and the people who run them. It is activated when (a) the main site or local systems are expected to be unavailable for more than two hours, (b) a function with a deadline inside 24 hours is at risk, or (c) a backup or recovery integrity problem is found. This event meets (a) and (b) at 08:00, so activation is immediate. The authorised decision maker is the Director of Operations as Incident Commander, with the Head of IT as deputy; I coordinate the plan and report to both. Escalation runs from me to the Incident Commander within 10 minutes and to the Principal within 30 minutes. Contact roles sit on an offline call-tree list; alternative communication uses Microsoft 365 over mobile internet, then phone and SMS. Situation reports go out hourly for 12 hours (status, impacts, actions, decisions needed, next update). If the Incident Commander cannot be reached within 10 minutes, the deputy takes the same authority, so activation never waits for one person, and every activation decision is logged with its time and rationale.

The four teams work under one command (slide 14). The continuity team (business coordinators, led by me) runs temporary services and learner support; the technical recovery team (two technicians, led by the Head of IT) executes the runbook; the Facilities Manager manages the alternate facility, power and safety clearance; and the incident response team, led by the Information Security Lead, stands by. They hold a 15-minute joint stand-up each hour and share one log. Incident response may pause any restoration step that touches suspect media, but the Incident Commander decides the speed-versus-safety trade-off. Incident response is triggered by any of: an unexpected logon to backup or directory administration, a failed integrity check with no clear technical cause, unexplained data changes, or payment records that contradict trusted external data. I then notify the Information Security Lead within 15 minutes, preserve logs and stop using the affected media until triage.

Table 10: RACI matrix (exactly one Accountable per activity; A/R = combined role)

Activity	Accountable	Responsible	Consulted	Informed
1. Plan activation and declaration	Incident Commander, Director of Operations (A/R)	BC/DR Coordinator	Head of IT; Facilities Manager	Principal; all staff
2. Temporary assessment delivery	Head of Academic Delivery (A/R)	Exams Officer; invigilators; BC coordinators	Head of IT; Information Security Lead	Incident Commander; learners
3. Backup validation	Head of IT (A/R)	Technician 1; Information Security Lead (independent check)	BC/DR Coordinator	Incident Commander
4. System restoration	Head of IT (A)	Technicians 1 and 2	Information Security Lead; function owners	Incident Commander; BC/DR Coordinator

Activity	Accountable	Responsible	Consulted	Informed
5. Stakeholder communication	Incident Commander (A)	Communications Lead; BC/DR Coordinator	Function owners; Information Security Lead	Learners; staff; providers
6. Incident response escalation and evidence	Information Security Lead (A/R)	Technicians (log preservation)	Incident Commander	Principal
7. Reconciliation of temporary records	BC/DR Coordinator (A)	Finance Manager; Admissions and Registry Manager; HR and Payroll Manager; Registrar	Head of IT	Incident Commander
8. Return to the primary site	Incident Commander (A)	Facilities Manager; Head of IT; function owners	Information Security Lead; Principal	All staff; learners

4.2 Temporary procedures for the 72-hour outage

Table 11 keeps each function at minimum service. The central constraint is examination capacity: of 120 candidates, 80 use their own devices and 40 need a centre workstation, but the alternate facility supplies only 25, a shortfall of 15. At 11:00 I seat 105 candidates (25 on workstations, 80 on own devices) and run a second sitting for the other 15 with an alternate paper, so no learner is excluded. Workstations go first to learners with accessibility needs and no alternative device. Integrity controls are photo-ID checks, invigilation (assumed 1:20, so six invigilators, A6), time-limited secure delivery and a log reconciled to the candidate list. Privacy controls limit temporary mailboxes and sheets to named staff, with deletion after verified transfer.

Table 11: Temporary procedures for F1 to F5

Function	Temporary procedure and approval	Records and later reconciliation	Minimum staff, equipment, supplier
F1	Secure Microsoft 365 assessment; Head of Academic Delivery approves the alternate paper and method. Accessibility adjustments (extra time, assistive tools) honoured under normal rules. Sitting 1 at 11:00 (105 candidates); sitting 2 for 15 later, using restored LMS01 only after owner acceptance.	Numbered attendance and submission log; scripts or exports reconciled to the candidate list and to LMS01 after restoration; marks entered only after acceptance.	Exams Officer; 6 invigilators; 1 technician on call; 25 workstations; mobile hotspots; Microsoft 365; alternate-facility provider.
F2	Payments continue through the provider; two-person maker-checker log; numbered manual receipts; Finance Manager approves; fees confirmed by 16:00 from the provider report.	Provider statement matched to FIN01 wallet records after restoration; every variance approved and logged.	Two finance officers; laptops; payment provider; bank.

Function	Temporary procedure and approval	Records and later reconciliation	Minimum staff, equipment, supplier
F3	Controlled admissions mailbox and form; reference numbers; Admissions and Registry Manager approves exceptions; 18:00 deadline kept; two-person count at close.	Batch entry into REG01 after restoration; counts and duplicates verified; temporary store deleted after transfer.	Two admissions staff; Microsoft 365; mobile internet.
F4	Use HR01 once restored (about 12:15). If it fails: last approved payroll plus variance sheet; HR and Finance managers both authorise; duplicate-payment check.	Run compared with the prior month; any manual changes posted to HR01 on return; errors corrected before the bank file is sent.	Payroll officer and approver; bank portal.
F5	Log requests; issue nothing from unverified copies; Registrar approves releases; access list retained.	Logged requests reconciled with FS01 after verification; permissions checked.	One records officer; FS01 (restored about 10:30).

4.3 DR runbook

Table 12 follows the technical dependency order and distinguishes restoring technology from resuming an acceptable business service: steps 3 to 8 are technically complete when their checks pass, but a function resumes only at step 9, when its owner signs acceptance and temporary records are reconciled. Rollback means returning to manual procedures and restoring from the previous valid checkpoint.

Table 12: DR runbook

#	Action	Prerequisite	Owner	Completion check	If the step fails
1	Declare activation; confirm no compromise indicated; assign roles	None	Incident Commander	Roles confirmed; first situation report sent	If unreachable within 10 min, Head of IT activates
2	Select backups: B01 + B02 (DB01), B03 (AUTH01), B04 (FS01)	Step 1	Head of IT	Catalogue shows unbroken chain to 07:00; hashes match	Use last unbroken point; use the 17:00 Day −3 full only after IR clearance
3	Restore recovery network (30 min)	Step 2	Technician 1	Connectivity and firewall tests pass	Mobile hotspots for Microsoft 365 work; escalate to site provider
4	Restore AUTH01 from B03 (30 min)	Step 3	Technician 1	Test logon works; object counts and time sync checked	Rebuild minimal directory for named users; re-issue credentials
5	Security and backup checks (30 min)	Step 4	Technician 1; Information Security Lead	Malware scan clean; hashes verified; backup administrator logons reviewed	Hold restoration; invoke incident response; use immutable copy

#	Action	Prerequisite	Owner	Completion check	If the step fails
6	Restore DB01: B01, then each incremental in order to 07:00 (90 min)	Step 5	Technician 1	Consistency check passes; last transaction time 07:00; record counts match	Restore to last good point; reconcile the gap
7	Restore FS01 from B04 (60 min), in parallel with step 6	Steps 4 and 5	Technician 2	File counts and hash sample match; permissions intact	Serve requests from verified Microsoft 365 copies
8	Restore LMS01 (45), FIN01 (30), REG01 (30), HR01 (45)	Step 6	Technicians 1 and 2	Logon, sample records and a test transaction per application	Keep that function on manual procedures; log defect
9	Business acceptance and reconciliation of temporary records	Step 8	Function owners; BC/DR Coordinator	Owner signs minimum-service acceptance; reconciliation complete	Keep manual procedures; extend support
10	Rollback (any failed validation after go-live)	Failure found	Head of IT	Access suspended; manual procedure resumed; prior checkpoint restored	Escalate to Incident Commander; consider immutable copy

4.4 Stakeholder communication

Table 13 lists who is told what, and the initial learner notice follows it. The notice is sent by 08:45 after approval by the Incident Commander.

Table 13: Stakeholder communication matrix

Stakeholder	Message	Channel	Owner	Timing
Learners (1,800)	Closure, exam arrangements, payment and admissions channels, update schedule	Email; SMS	Communications Lead (approved by Incident Commander)	By 08:45; then every 4 h
Exam candidates (120)	Sitting time, venue, photo ID, workstation request	Email; SMS	Exams Officer	By 08:45; replies by 09:30
Staff (90)	Roles, safety, working arrangements, payroll status	Teams; SMS	BC/DR Coordinator	By 08:30; hourly to teams
Applicants	Temporary channel and 18:00 deadline	Email; website	Admissions and Registry Manager	By 09:30
Payment provider and bank	Outage, reconciliation contact, payroll deadline	Phone; email	Finance Manager	By 09:00

Stakeholder	Message	Channel	Owner	Timing
Alternate facility and backup provider	Activation, access, restoration support	Phone	Facilities Manager; Head of IT	By 08:15
Principal and governing board	Impacts, decisions needed	Phone; report	Incident Commander	Within 30 min; then every 4 h

Initial learner notice

Subject: Northbridge Skills Institute: temporary campus closure today

Dear learner,

Flooding and a power failure have closed our main campus from 08:00 today. We expect it to stay closed for about 72 hours, so please do not travel to the campus. Your safety comes first.

Today's 11:00 examination will go ahead. If you are a registered candidate, check your Northbridge email for your sitting time, venue and instructions, and bring photo identification. If you need a centre workstation, reply to this email by 09:30 so we can allocate your place. Some candidates may sit a later session today.

You can still pay fees through the usual payment provider. Admissions remain open until 18:00 today through admissions-temp@northbridge.example. Your records are protected and will be updated when our systems return.

We will send updates by email and SMS at least every four hours. If you need help, contact the learner support line.

Northbridge Skills Institute Continuity Team

4.5 Safe return and sign-off

Return to the primary site needs three things. Facilities gives written clearance (structural, electrical and water safety, with power stable for 24 hours). Technical readiness is confirmed: equipment is inspected and tested, the network is verified and the Information Security Lead confirms no unresolved compromise. Each function owner completes business validation. Records created during the outage move under control: I fix a cut-over time, freeze temporary stores, export them, have two people verify counts and totals, import them into the primary systems and reconcile payments, applications and requests. The Incident Commander gives final sign-off only when Facilities clearance and business validation are both documented, and a lessons-learned review within 10 working days feeds the corrective actions in Task 5.

Task 5: Recovery schedule and exercise

Table 14 gives the baseline schedule using the current warm-site timings and respecting the two-technician limit and prerequisites. It is a modelled baseline, not a live test. The chain is sequential until DB01 finishes; only FS01 can run beside DB01, on the second technician.

Table 14: Modelled baseline recovery schedule (Day 0)

Action	Technician	Start	Finish	Prerequisite
Recovery network (30 min)	T1	08:00	08:30	None
AUTH01 restoration (30 min)	T1	08:30	09:00	Recovery network
Security and backup checks (30 min)	T1	09:00	09:30	AUTH01 restoration
DB01 restoration (90 min)	T1	09:30	11:00	Security and backup checks
FS01 restoration (60 min)	T2	09:30	10:30	AUTH01 restoration; security and backup checks
LMS01 restoration (45 min)	T1	11:00	11:45	DB01 restoration
FIN01 restoration (30 min)	T2	11:00	11:30	DB01 restoration
HR01 restoration (45 min)	T2	11:30	12:15	DB01 restoration
REG01 restoration (30 min)	T1	11:45	12:15	DB01 restoration

The assessment service is restored at 11:45, the earliest possible time. The controlling dependency path is recovery network (30) → AUTH01 (30) → security and backup checks (30) → DB01 (90) → LMS01 (45) = 225 minutes. All five functions finish at 12:15: after 11:00, four actions totalling 150 minutes share two technicians, so 75 minutes is the minimum, and my split achieves it. Because LMS01 finishes 45 minutes after the examination starts, F1 needs the continuity workaround from Task 4 even though restoration succeeds.

Table 15: Elapsed recovery time against RTO and MTD, and checkpoint against RPO (modelled baseline)

Function	Finish	Elapsed from 08:00	RTO	MTD	RTO met?	Checkpoint age vs RPO	RPO met?
F1	11:45	3 h 45 min (225 min)	4 h	8 h	Yes (15 min margin)	1 h vs 1 h	Yes (no headroom)
F2	11:30	3 h 30 min (210 min)	4 h	8 h	Yes (30 min margin)	1 h vs 1 h	Yes (no headroom)
F3	12:15	4 h 15 min (255 min)	5 h	10 h	Yes (45 min margin)	1 h vs 1 h	Yes (no headroom)
F4	12:15	4 h 15 min (255 min)	6 h	24 h	Yes (1 h 45 min)	1 h vs 4 h	Yes

Function	Finish	Elapsed from 08:00	RTO	MTD	RTO met?	Checkpoint age vs RPO	RPO met?
F5	10:30	2 h 30 min (150 min)	12 h	48 h	Yes (9 h 30 min)	9 h vs 8 h	**No** (1 h gap)
AUTH01	09:00	1 h (60 min)	2 h	8 h	Yes (1 h)	8 h vs 4 h	**No** (4 h gap)

Every function meets its RTO and sits well inside its MTD, but F1 has only 15 minutes of margin. F5 and AUTH01 miss their RPOs because their checkpoints are older than the targets. Only F1 needs a continuity workaround to meet its deadline, since the 11:00 examination falls before 11:45. F2, F3 and F4 meet their deadlines after restoration but still need reconciliation of temporary records, because restored data stops at 07:00.

The 45-minute tabletop in Table 16 is a separate stress scenario and does not change the baseline. I simulate every role myself: Incident Commander, Head of IT, Facilities Manager, Academic Lead, Finance Manager and Information Security Lead. Finish times in the stress case are planning estimates.

Table 16: Tabletop exercise log (stress scenario, separate from baseline)

Min	Inject	Decision and revised action	Owner	Rationale	Effect on service or recovery
10	Closure extends from 72 hours to five days	Re-declare an extended outage. Extend alternate-facility booking to five days (to Saturday 08:00); set a shift rota for the two technicians; request a supplier engineer on call; reissue learner notice; move temporary procedures from 72 hours to five days.	Incident Commander; Facilities Manager	A longer outage raises fatigue, cost and the risk of manual-process errors.	Baseline restoration unchanged; continuity workload and cost rise; learners and staff re-notified.
20	Alternate facility offers only 15 working workstations	Allocate 15 workstations to the highest-need candidates for sitting 1, with the 80 own-device candidates at 11:00 (95 in total). Run sitting 2 for the other 25 on the same 15 workstations plus 10 loaned laptops or a paper fallback, after LMS01 acceptance. Use an alternate paper.	Head of Academic Delivery; Facilities Manager	Avoids excluding candidates or overcrowding; the shortfall rises from 15 to 25.	25 candidates move to a later sitting; extra invigilation; examination still held on Day 0; schedule unchanged.

Min	Inject	Decision and revised action	Owner	Rationale	Effect on service or recovery
30	A DB01 incremental fails its integrity check; unexpected backup administrator logon	Treat as possible compromise and trigger incident response. Information Security Lead takes control; disable the shared administrator identity and issue break-glass credentials; isolate the backup console and preserve logs. Restore DB01 to the last unbroken chain point (no earlier than B01, 23:00 Day −1). Use the 17:00 Day −3 full (63 h old) only after IR clearance. Restore FS01 only after B04 hash verification.	Incident Commander; Information Security Lead; Head of IT	An unexplained integrity failure plus an unexpected administrator logon meets the trigger; restoring unverified media could spread compromise.	DB01 starts about 10:30 (60 min triage while T2 restores FS01) and ends 12:00. F1 12:45, F2 12:30, F3 and F4 13:15. F1 and F2 exceed RTO (4 h 45 min, 4 h 30 min) but stay inside MTD; F3 is 15 min over RTO (5 h 15 min). Data loss rises to as much as 9 h.
40	Payment provider reports successful payments missing from restored finance records	Treat the provider report as authoritative external evidence. Hold new receipts until reconciled; match the provider list from 07:00 to now against FIN01; enter missing payments as maker-checker adjustments. A missing payment dated before 07:00 is an integrity failure and goes to incident response. Confirm fees by 16:00 from the reconciled list.	Finance Manager (R); BC/DR Coordinator (A)	Provider records are independent. A post-07:00 gap is expected with a 1 h RPO; a pre-07:00 gap is not.	16:00 confirmation remains feasible if reconciled by about 15:00; about 2 extra staff-hours; learners protected from double charges.

Table 17: Corrective actions

#	Corrective action	Priority	Owner	Target date	Measurable completion evidence
1	Separate the shared production and network backup administrator identity: dedicated backup account with MFA, no standing delete rights, logon alerts	Critical	Head of IT	20 Nov 2026	Access review shows no shared administrator accounts; alert test recorded; production credentials cannot delete backups in a test
2	Document and test the offsite provider’s access independence	Critical	Head of IT with Procurement	18 Dec 2026	Signed provider statement; test shows production credentials cannot read, alter or delete offsite or immutable copies
3	Implement the Table 6 schedule: 30-minute DB01 incrementals, 4-hour AUTH01 snapshots, weekly immutable copy, daily chain check	High	Head of IT	29 Jan 2027	Four consecutive weeks with 100% chains complete; monthly restore test shows DB01 RPO ≤ 30 min and AUTH01 ≤ 4 h

#	Corrective action	Priority	Owner	Target date	Measurable completion evidence
4	Secure alternate capacity for 40 centre candidates plus staff, or a loaned-device pool	High	Director of Operations (Facilities)	29 Jan 2027	Signed agreement listing at least 40 workstations, or a pool of 15 loaned devices confirmed by on-site count
5	Build and approve an assessment continuity pack: alternate papers, Microsoft 365 delivery, paper fallback	High	Head of Academic Delivery	26 Feb 2027	Academic Board approval minute; pilot with at least 20 learners, log reconciled 100% to candidate list
6	Add a third trained recovery person or supplier call-out; run a timed restore rehearsal	Medium	Head of IT	31 Mar 2027	Named person completes runbook steps 3 to 8 within 225 min; supplier call-out SLA of 2 h or better
7	Prepare a payment reconciliation procedure with provider report access	Medium	Finance Manager	29 Jan 2027	Dry run reconciles a 100-transaction sample within 60 min with zero unexplained variances

These actions come from the baseline analysis and the tabletop, and the KPIs in Task 6 track their closure.

Task 6: Governance and performance measures

I own the plan day to day as BC/DR Coordinator. The Executive Steering Committee, chaired by the Principal, approves the plan, the budget and residual-risk acceptance, as slide 12 describes, and each function owner owns their temporary procedure. Version control uses a document ID, version number, change log and owner and approver signatures in one controlled repository with restricted edit rights. Distribution follows a named list: the Incident Commander, deputy, coordinator, technicians and function owners hold current copies, with encrypted offline and printed copies at the alternate facility; superseded copies are withdrawn.

The testing calendar (Table 18) applies the slide 11 methods, scaled to risk, with monthly DB01 restore tests because DB01 supports four functions. Reviews are also triggered by system or supplier changes, a failed test, a real incident, an audit finding, a change of site or a flood warning. Role-holders and deputies are trained within 30 days of appointment and annually; technicians drill the runbook quarterly.

Table 18: Risk-based testing calendar

Test	Frequency and timing	Purpose	Owner
Backup restore tests	DB01 monthly; AUTH01 and FS01 quarterly	Prove integrity, chain completeness, RPO and usability	Head of IT
Tabletop exercise	Quarterly; one before the seasonal flood period	Decision-making and role clarity	BC/DR Coordinator
Walkthrough	Semi-annually	Validate runbook and RACI	BC/DR Coordinator
Simulation (timed)	Annually	Rehearse the Table 14 schedule under time pressure	Incident Commander
Parallel test at the warm site	Annually, before the flood period	Validate supplier timings with our data	Head of IT
Full interruption test	Not before Level 3 is reached; then every 2 to 3 years	Complete validation	Executive Sponsor decides

Flood consideration: the 2024 climate amendment to ISO 22301 adds a requirement to determine whether climate change is a relevant issue when understanding the organisation's context (ISO, 2024). For Northbridge this means assessing flood exposure of the main site, the alternate facility and the offsite data location so that no two share a flood zone or power grid, raising critical equipment above likely water levels, and timing the annual parallel test and a tabletop before the seasonal flood period, which slide 5 treats as a timing factor in the BIA.

Table 19 maps my outputs to the seven NIST SP 800-34 planning steps (slide 4) and to the ISO 22301 clause themes (slide 9) as an educational alignment, with no claim of certification.

Table 19: Mapping of outputs to NIST SP 800-34 steps and ISO 22301 themes

Framework element	Output in this report
NIST step 1: Contingency planning policy	Plan ownership, scope, activation and approval (Tasks 4 and 6)
NIST step 2: Business Impact Analysis	Table 1, Table 2 and Table 3 (Task 1)
NIST step 3: Preventive controls	Backup design, separate administrator identity, flood siting (Table 6; Task 6)
NIST step 4: Contingency strategies	Option matrix, budget and recommendation (Task 3)
NIST step 5: Information system contingency plan	Plan, RACI, temporary procedures and runbook (Task 4)
NIST step 6: Testing, training and exercises	Baseline schedule, tabletop and testing calendar (Tasks 5 and 6)
NIST step 7: Plan maintenance	Version control, review triggers and corrective-action closure (Tasks 5 and 6)
ISO clause 4: Context	Flood and climate consideration; stakeholder matrix
ISO clause 5: Leadership	Steering Committee, Incident Commander authority, risk acceptance
ISO clause 6: Planning	RTO, RPO and MTD objectives; KPI targets
ISO clause 7: Support	Training, resources, communication and documented information
ISO clause 8: Operation	BIA, strategy, plan, runbook and exercises
ISO clause 9: Performance	KPI dashboard and exercise results
ISO clause 10: Improvement	Corrective-action register

Table 20: Performance dashboard: KPIs and modelled baseline

KPI	Formula or method	Data source	Target	Reporting	Owner	Modelled baseline
Recovery time attainment	(Functions restored within RTO ÷ functions tested) × 100; also actual vs target RTO	Exercise log; restore timer	100% for F1 to F3; at least 90% overall	After each test; quarterly report	Head of IT	5 of 5 = 100% (modelled; F1 margin 15 min)
Recovery point attainment	Disruption time − restored checkpoint time, compared with RPO	Backup catalogue; restore test	At or under RPO for every item	Monthly	Head of IT	4 of 6 met; F5 +1 h, AUTH01 +4 h
Restoration-test coverage	(Systems with a successful restore in the last 90 days ÷ in-scope systems) × 100	Test log	100% (DB01 monthly)	Monthly	Head of IT	Not evidenced (integrity checks only)
Corrective-action closure	(Actions closed on time with evidence ÷ actions due) × 100	Action tracker	At least 90%	Monthly	BC/DR Coordinator	0 of 7 closed (new)

KPI	Formula or method	Data source	Target	Reporting	Owner	Modelled baseline
Training completion	(Role-holders and deputies trained in last 12 months ÷ total) × 100	Training register	100%	Quarterly	BC/DR Coordinator	Not recorded
Backup chain integrity	(Chains passing integrity and completeness checks ÷ chains checked) × 100	Backup report	100%	Weekly	Head of IT	Baseline drill: B01 to B04 passed; independence unverified

On the slide 13 model I assess Northbridge at Level 2 (Developing): backups, a warm-site drill and basic plans exist, but testing is not regular, the backup administrator identity is shared and supplier claims are unvalidated. A realistic 12-month target is Level 3 (Defined): a documented plan, regular testing, clear governance and consistent recovery. Level 4 needs data-driven measurement and enterprise risk integration, which is not realistic within a year. Evidence for Level 3 is closure of all seven corrective actions, four tabletops and one annual simulation completed, and the KPIs reported to the Steering Committee each quarter.

In summary, the modelled baseline shows that a Warm recovery can restore all five functions by 12:15 and meet every RTO, but it also shows three weaknesses that the budget alone does not fix: the 11:00 examination cannot be saved by restoration, the AUTH01 and FS01 checkpoints are older than their RPOs, and the shared administrator identity and undocumented provider controls leave recoverability unproven. My recommendation therefore pairs the NGN 2,100,000 Warm package with the continuity procedures in Task 4 and the seven corrective actions in Task 5. The Steering Committee should approve the spend only after the supplier timings and backup independence have been validated, and should review the KPI dashboard each quarter against the Level 3 target.

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NIST (2010) SP 800-34 Revision 1: Contingency Planning Guide for Federal Information Systems. National Institute of Standards and Technology. Available at: https://csrc.nist.gov/pubs/sp/800/34/r1/upd1/final

Appendix A: Assumptions and unverified items

ID	Assumption or unverified item	Used in	Status
A1	The number of teaching staff, applicants and record requesters is not supplied.	Task 1	Assumption
A2	Little business activity occurs between 07:00 and 08:00 on Day 0.	Task 2	Assumption
A3	Approved examination papers and the candidate list are held in Microsoft 365, and the Academic Board permits secure Microsoft 365 delivery of the assessment.	Tasks 1 and 4	Unverified
A4	B01 to B04 pass integrity checks and the incremental chain is contiguous (stated for the baseline drill only).	Tasks 2 and 5	Unverified
A5	The offsite provider's copies are independent of our administrator identity; the 17:00 Day −3 disconnected backup can be restored.	Tasks 2 and 5	Unverified
A6	Invigilation ratio of 1:20, giving six invigilators for 105 candidates.	Task 4	Assumption
A7	Technicians, coordinators and function owners have laptops, mobile data and Microsoft 365 access, and an SMS service exists.	Task 4	Assumption
A8	The named roles (Principal, Director of Operations, Head of IT, Information Security Lead and others) exist and can act as deputies.	Tasks 4 and 6	Assumption
A9	Package timings and data-loss windows are supplier estimates; the NGN 3,000,000 limit is inclusive.	Task 3	Assumption

Appendix B: AI assistance acknowledgement

I acknowledge that I used an AI assistant (Claude, by Anthropic) to help structure and draft parts of this report. I reviewed the content, checked the calculations against the case data, adapted the decisions to the scenario, and I take responsibility for the final submission. All scenario data are fictional.